User Manual

Local monitoring and control interface for ReSPR devices

ReSPR IRIS

INTEGRATED REMOTE INTELLIGENCE SYSTEM

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1. Introduction

ReSPR IRIS HMI is the local interface used to monitor and control enabled ReSPR devices connected through a mesh communication network.

From the HMI touch interface, the user can

- view device status,
- turn units on and off,
- program timers,
- check warnings and alarms,
- review connectivity data,
- record an NCC™ cell replacement
- and set preventive service dates.

The HMI is designed to make day-to-day monitoring of the installation easier. It does not replace physical checks of power supply, installation, coverage, service, or the actual condition of the device. The information shown depends on the data received from each node through the mesh network.

> **Important**: if a unit is not communicating correctly with the HMI, the screen may not reflect its actual status at that time. If you have any operational doubts, also inspect the device physically.

2. Scope of this manual

This manual describes the user-visible operation of the HMI for the end user. Device names, locations, models, and the total number of devices may vary depending on the specific configuration of each installation.

In the current reference configuration, the HMI can display up to 24 service units (nodes), organized into pages of 6 units. It can also monitor an infrastructure or repeater node when one is part of the installation.

The text visible on screen may vary slightly depending on the installed language, the final project configuration, or future updates validated by ReSPR.

3. System overview

The HMI displays monitoring information and allows actions to be performed on the configured devices.

The main information available is:

- devices configured in the project;
- number of online units compared with the total configured;
- warnings and alarms;
- connection status of each unit;
- NCC™ cell/lamp status;
- active, pending, or unscheduled timers;
- estimated remaining life of the NCC™ cell;
- next preventive service;
- technical data for the selected unit, such as voltage, current, relay status, signal, reconnections, and outages.

The main actions available are:

- turn on one or more units;
- turn off one or more units;
- program timers;
- cancel timers;
- record an NCC™ cell/lamp replacement;
- set the next preventive service;
- restart the HMI;
- perform a software restart of a selected node.

> **Important**: recording a cell replacement in the HMI should only be done once the NCC™ cell or lamp has been physically replaced.

4. Main screens

The HMI is organized into three main areas:

Screen	Main use
Screen 0 - Startup / connection	Initial device search and startup help.
Screen 1 - Dashboard	General monitoring of all devices and actions on multiple units.
Screen 2 - Device details	View and individually control a specific unit.

5. Screen 0 - Startup and connection

When starting up, the HMI displays an initial connection screen. This screen indicates that the system is searching for available ReSPR units on the mesh network.

It may display messages similar to:

- "Searching for online devices...";
- "The connection is taking longer than expected...";
- "Retrying device detection (...)."

The initial screen remains active until the HMI detects at least one valid unit with stable communication.

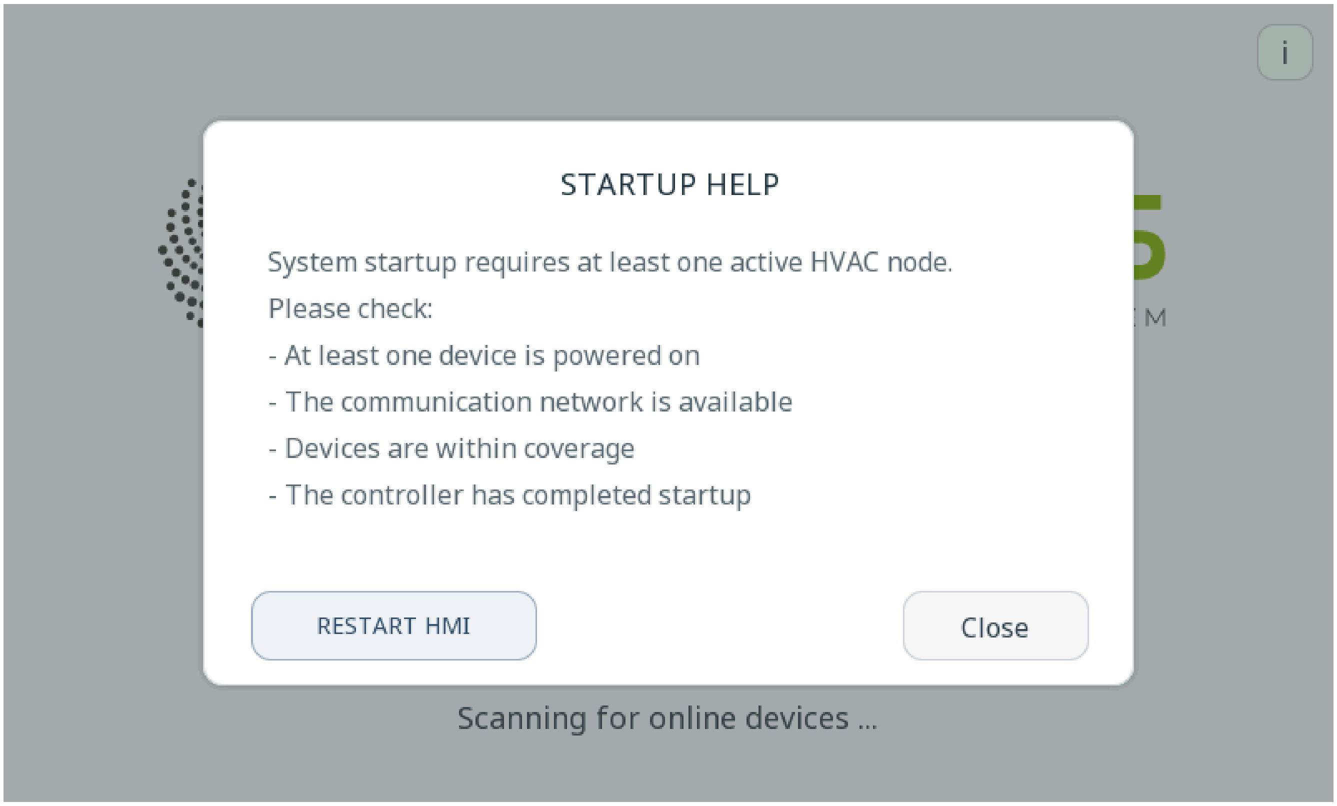
[SCREENSHOT 01 - Screen 0: searching for online units]

Startup help

The initial screen includes a help button. This help guides the user through basic checks:

- check that at least one unit is powered on;
- check that the communication network is available;
- check that the units are within coverage;
- wait for the controller to complete startup.

The **Restart HMI** option may also be available from this help screen.

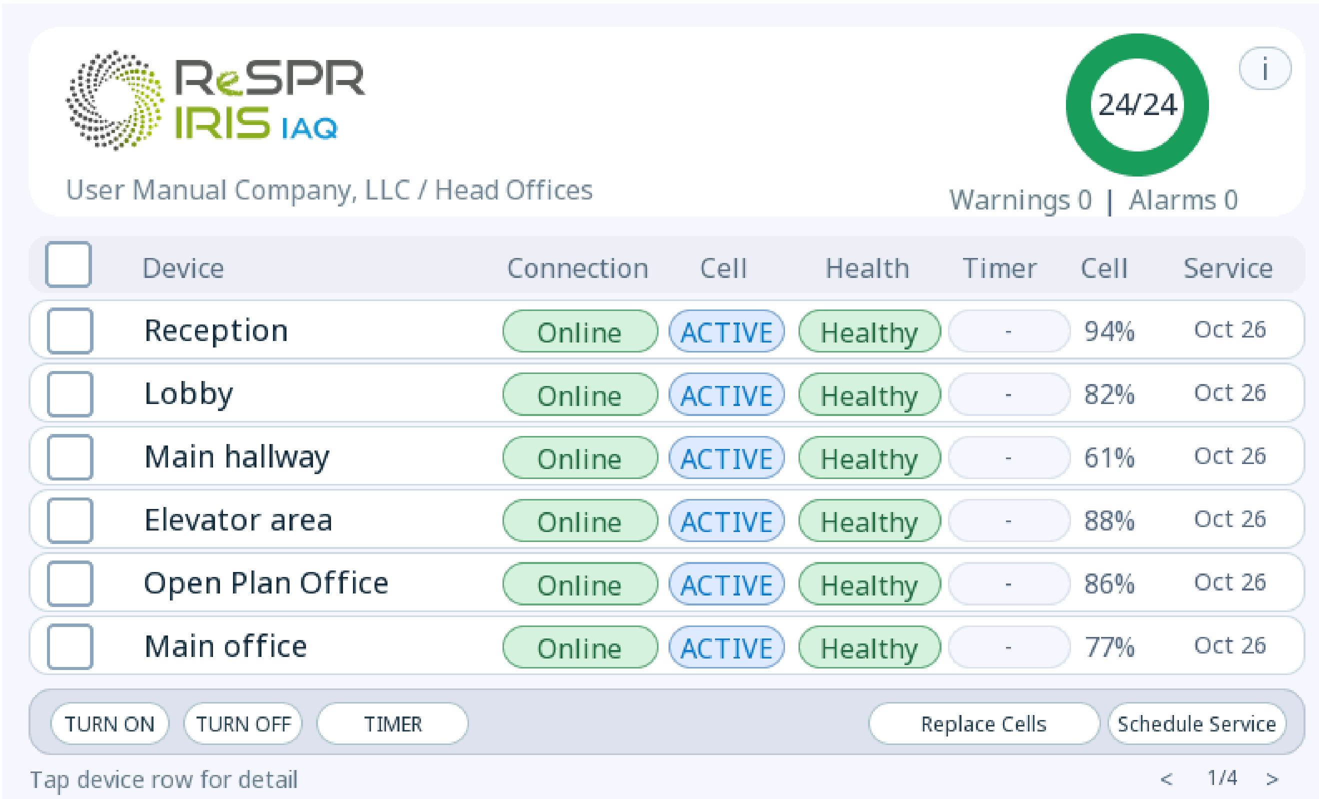


[SCREENSHOT 02 - Startup help with Restart HMI option]

Important: Restart HMI restarts only the interface. It does not restart the devices, erase node counters, or modify the physical status of the devices.

6. Screen 1 - Main dashboard

Screen 1 is the main monitoring view. It summarizes the status of all configured devices and allows actions to be run on one or more selected units.

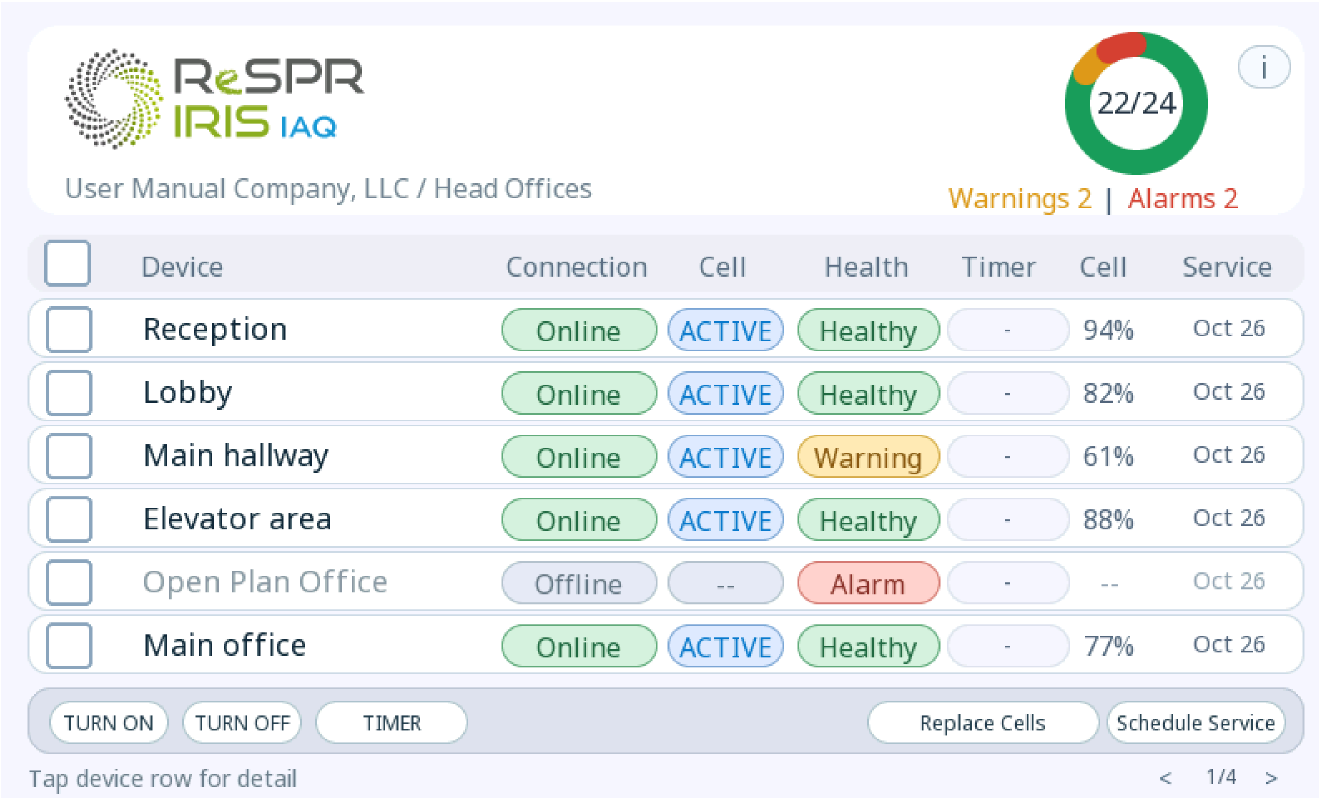


[SCREENSHOT 03 - Dashboard in normal status]

Online devices counter

The header shows a counter in **online/total** format, for example **1/24**. This counter indicates how many units have recent communication with the HMI compared with the total number of configured units.

The summary ring visually distributes the installation across healthy statuses, warnings, alarms, and offline devices. It also shows specific counters for **Warnings and Alarms**.



[SCREENSHOT 04 - Dashboard with warnings and alarms]

General dashboard statuses

The dashboard distinguishes the following statuses:

Status	General meaning
Healthy	The device does not have a relevant warning or alarm condition.
Warning	There is a condition that requires attention, but it is not treated as a critical fault.
Alarm	There is a priority condition that requires review.
Offline	The HMI is not receiving a recent status from the device.

An offline unit may be shown as an **alarm** if the last confirmed status was **on**. This logic avoids treating a loss of communication during devices operation as neutral.

Device rows

Each row represents one unit. The row shows:

- device name;
- connection;
- cell/lamp status;
- health;
- timer;
- remaining life;
- next service.

Touching the main area of a row opens **Screen 2 - Device details**. Touching a row's selection box selects or deselects that unit for bulk actions.

Connection column

The **Connection** column indicates whether the unit is **Online** or **Offline**.

In the current configuration, the HMI considers a status recent for approximately 20 seconds after the last reception. If the unit has never sent a status, or if too much time has passed since the last status received, it is shown as offline.

Cell / lamp column

The **Cell** column summarizes the status of the NCC™ cell or lamp.

Visible status	Meaning
OFF	No power applied to the cell, no fresh status, or device inactive.
Startup	The device has just been turned on and the HMI is waiting for an NCC™ current reading.
ON	Cell active and current within the expected range.
FAULT	The relay is on, but sufficient current is not detected.

More specific statuses may appear on the detail screen, such as **STARTING**, **ACTIVE**, **LOW**, **FAULT**, or **INACTIVE**.

Health column

The **Health** column summarizes the condition of the unit.

Visible status	Meaning
Good	Normal operation or non-critical condition.
Warning	Current below target, pending timer, or another notification that requires review.
Alarm	Low bus voltage, relay on without active NCC™ current, or device offline when the last confirmed status was ON.
Offline	No recent status and no condition that elevates the device to warning or alarm.

Timer column

The **Timer** column shows the timer status for the unit:

Visible status	Meaning
-	No timer is programmed.
Time remaining	If the timer is programmed to turn off, the remaining time will be shown in compact format on a red background. The remaining time for a unit turn-on schedule will be shown on a green background.
Pending	The timer should already have run, but the HMI has not yet received confirmation from the unit.

Visible timers are updated even when new statuses are not arriving from the unit (node).

Life column

The **Life** column shows the estimated percentage of remaining NCC™ cell life. This calculation is based on the lamp hours received from the node and the nominal life of the device profile.

If a cell replacement action is pending or synchronizing, statuses such as the following may appear:

Visible status	Meaning
Queued	The cell replacement action is waiting to be sent.
Sync	The command has been sent and the HMI is waiting for confirmation.
Done	The replacement was recently confirmed.

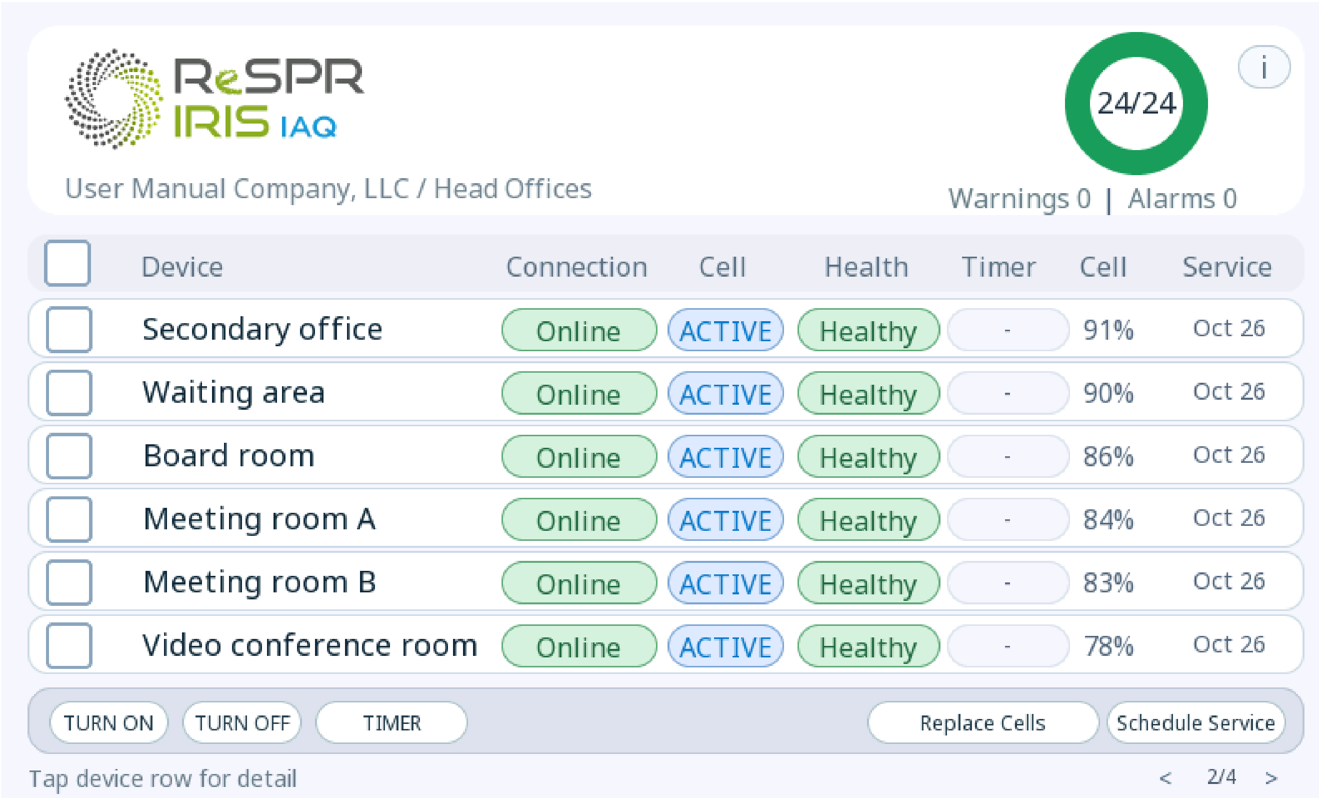
Service column

The **Service** column shows the next preventive service in short month and year format, or - if no date has been defined.

The HMI can use a date configured in the project or a date/interval set from the interface itself. Dates set from the HMI are saved locally.

Pagination

The dashboard shows 6 units per page. The < and > controls in the lower-right footer allow you to change pages. The page label shows the current number and the total, for example **1/4**.



[SCREENSHOT 05 - Dashboard pagination]

Important: when changing pages, the current selection is cleared.

Multiple selection

Each row includes a selection box. The selector in the table header selects or deselects all units visible on the current page.

Multiple selection applies only to the selected visible units on the current page. When changing pages, the selection is cleared.

Actions available from the dashboard

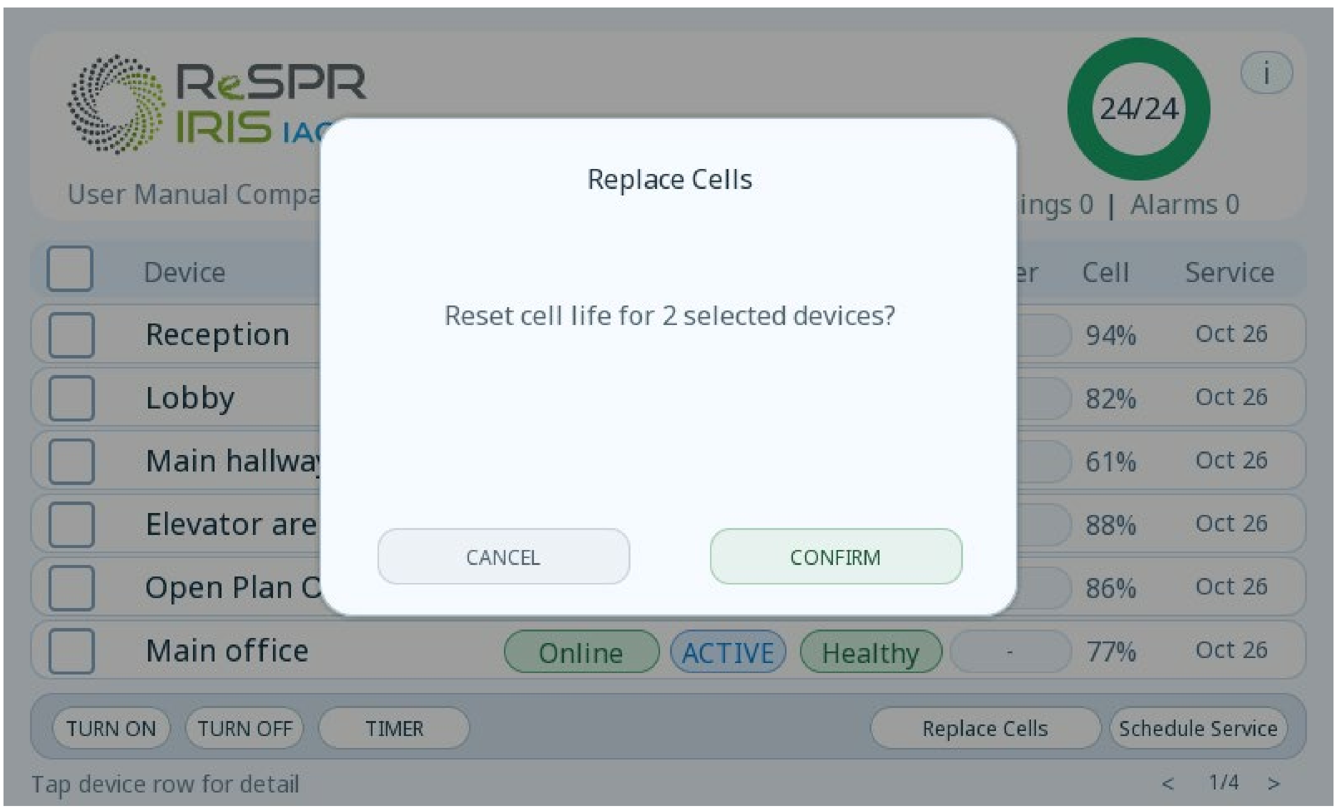
The following actions are available in the lower band of the dashboard:

Action	Function
TURN ON	Sends an ON command to the selected units.
TURN OFF	Sends an OFF command to the selected units.
TIMER	Opens timer programming for the selected units.
REPLACE CELLS	Records the NCC™ cell/lamp replacement for the selected units.
SCHEDULE SERVICE	Allows the month, year, and service interval to be set for the selected units.

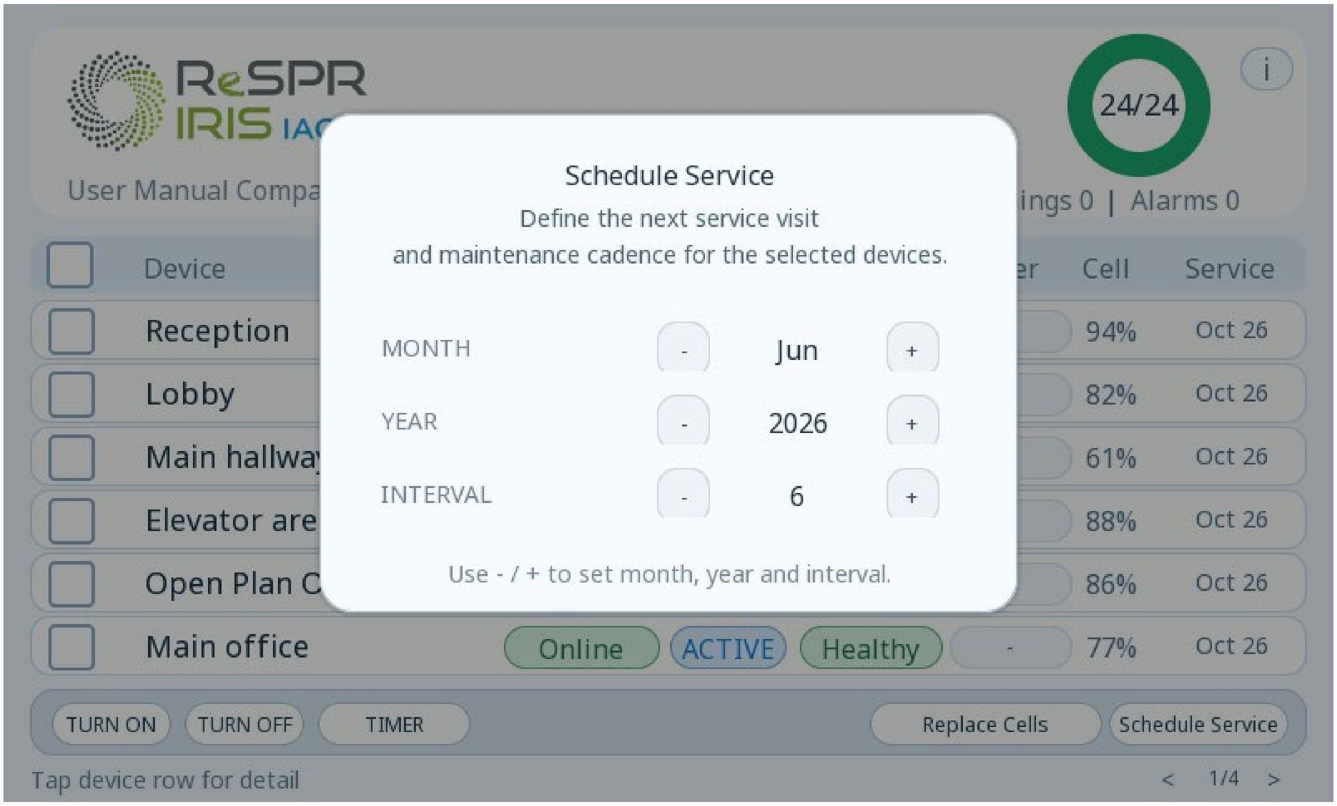
If only one unit is selected and **TURN ON** or **TURN OFF** is pressed, the HMI sends the action directly. If several units are selected, the HMI asks for confirmation.

For **REPLACE CELLS**, if any selected unit is offline, the action is queued for that unit. When the unit comes back online, the HMI will try to send the cell counter reset and wait for confirmation from the node.

For **SCHEDULE SERVICE**, the user adjusts the month, year, and interval with the - and + buttons. The user then saves and confirms a second summary screen before applying the change.



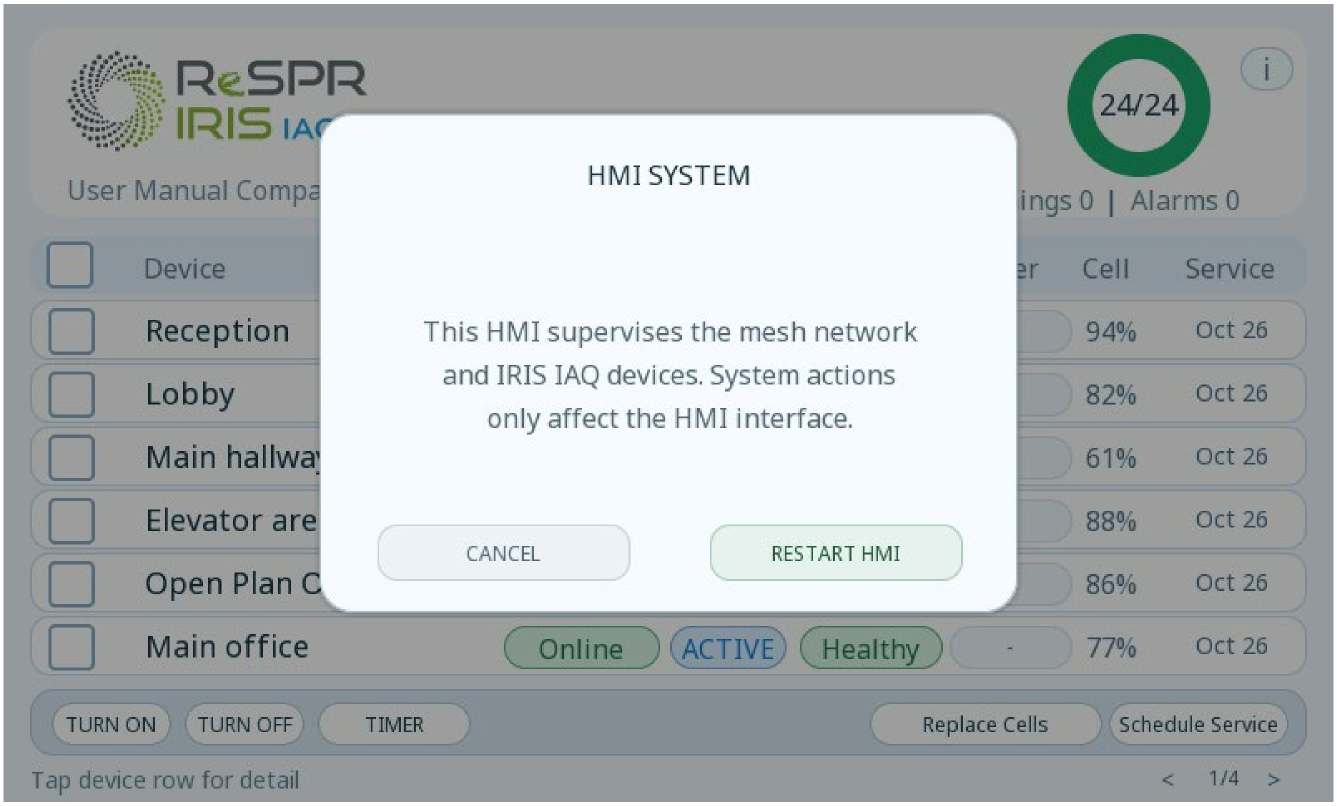
[SCREENSHOT 06 - Service modal]



[SCREENSHOT 06 6_BIS - Cell replacement modal]

HMI system information

The dashboard header includes an HMI system i button. This button opens information about the interface and allows the HMI to be restarted.

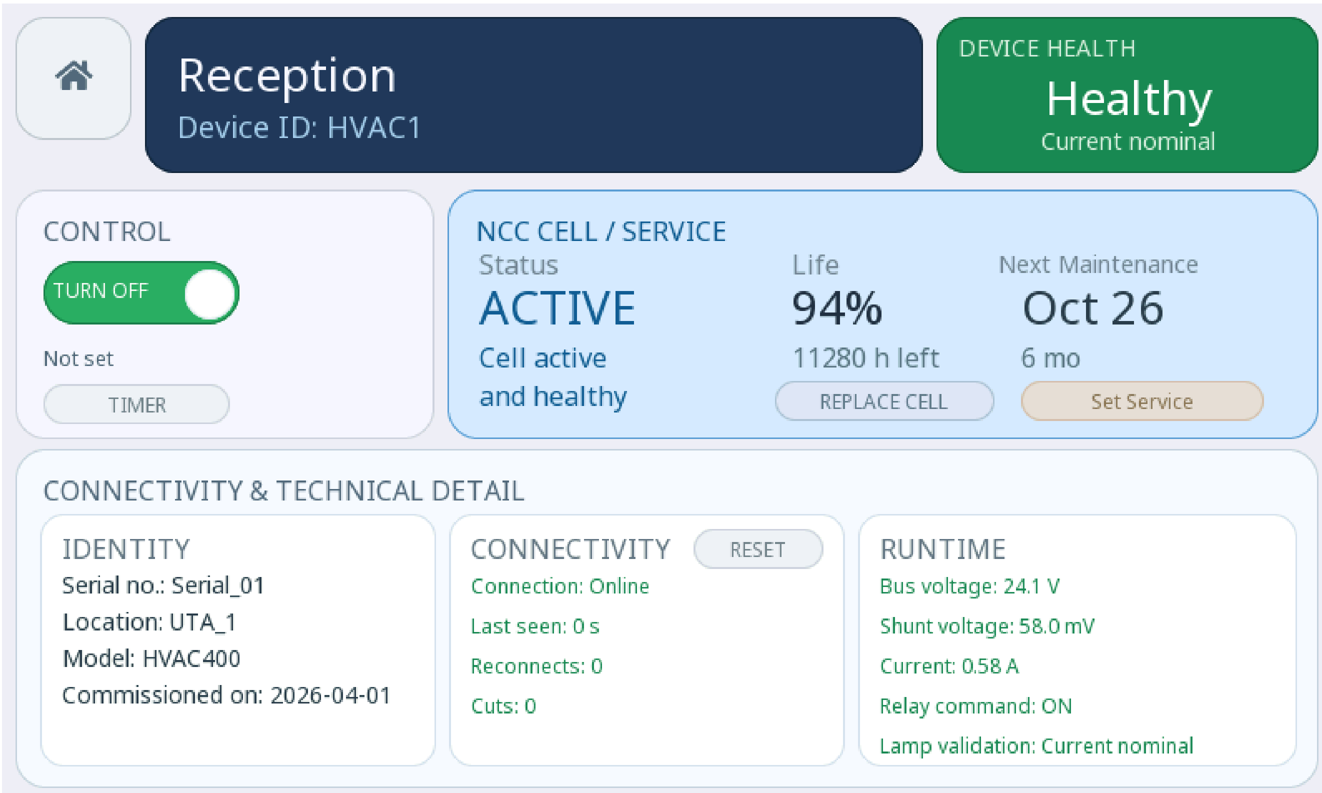


[SCREENSHOT 07 - HMI System modal from i button]

Important: restarting the HMI does not modify the nodes or erase their counters.

7. Screen 2 - Device details

Screen 2 opens when a dashboard row is touched. It shows detailed information for the selected unit and allows that unit to be operated individually.

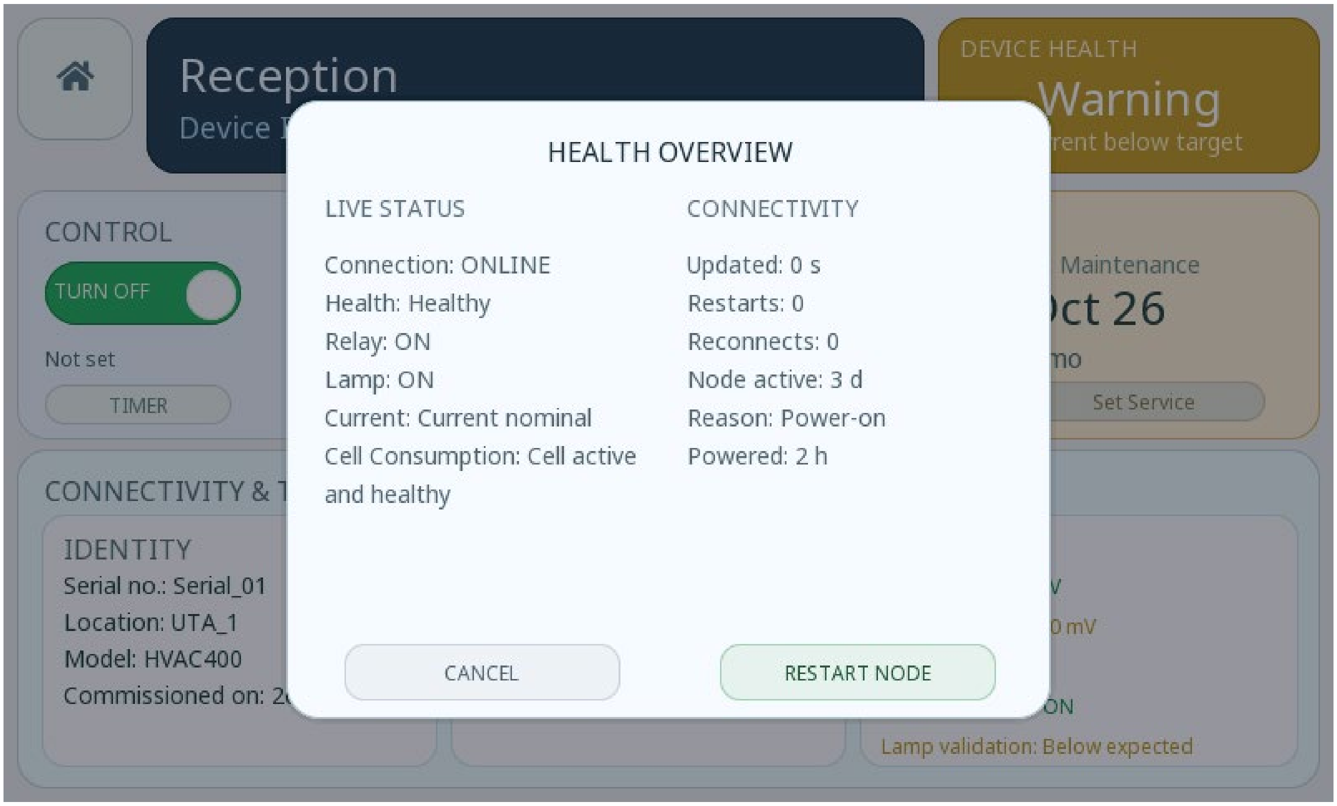


[SCREENSHOT 08 - Device details in healthy status]

General device status

The top section shows the identity of the unit and a general health panel. This panel summarizes device status based on the current connection, current/cell status, the presence of alarms, and timer status.

Touching the health panel opens a summary with live status and connectivity information.



[SCREENSHOT 09 - Screen 2 health summary]

ON/OFF control

The control panel shows the relay status and a touch control for switching between **ON** and **OFF**.

- If the unit is on, the control allows it to be turned off.
- If the unit is off, the control allows it to be turned on.

The action is sent to the active node over the mesh network. Final visual confirmation depends on the node sending an updated status to the HMI.

Device timer

Screen 2 shows the device timer summary:

Visible status	Meaning
Not set	No timer is active.
Turns on in X	The unit will turn on when the specified time has elapsed.
Turns off in X	The unit will turn off when the specified time has elapsed.
Pending ON	The turn-on command has already expired, but confirmation is missing.
Pending OFF	The turn-off command has already expired, but confirmation is missing.
ON now / OFF now	The target already appears confirmed in the received status when this text is available in the final version.

The **TIMER** button opens the same timer programming used from the dashboard, but applied only to the selected unit. More in section 8. Timer.

NCC™ cell and service

Screen 2 shows:

- NCC™ cell/lamp status;
- remaining life percentage;
- estimated remaining hours;
- next service;
- service interval, if defined.

Available actions:

Action	Use
REPLACE CELL	Records the cell/lamp replacement for the selected unit.
Set Service	Allows the next service date and interval to be set.

Important: **REPLACE CELL** resets the cell/lamp hour counter only after user confirmation and node confirmation. It must be used only when the NCC™ cell or lamp has been physically replaced.

Connectivity

The connectivity section shows:

- connection status;
- time seen online by the HMI;
- last signal received;
- reconnections;
- outages.

The **RESET** button, within the connectivity section**, resets the outage, reconnection, and active-time counters for the selected unit**. After confirmation, the HMI saves the change and restarts the interface.

> **Important**: this reset affects the connectivity counters visible for that unit in the HMI. It should not be confused with restarting the node or changing the cell.

Technical details

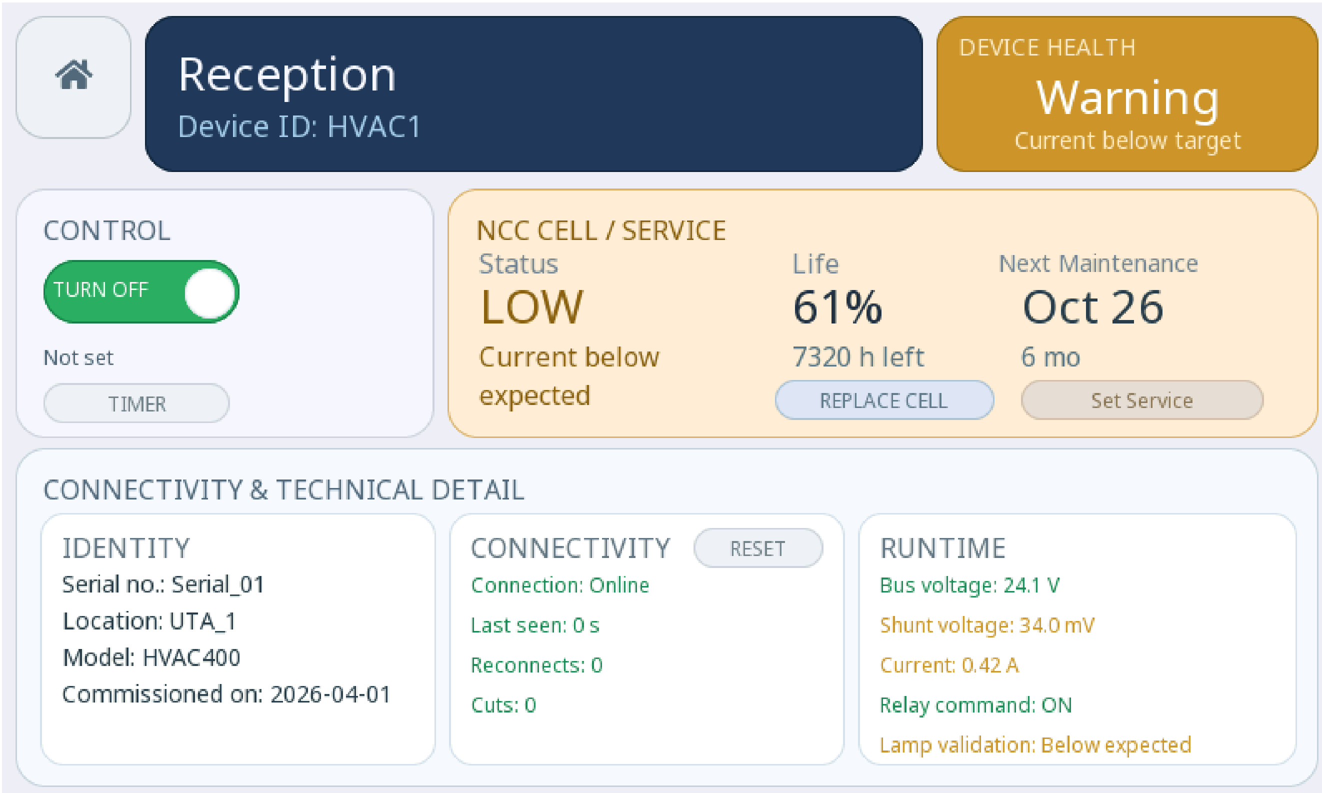
Screen 2 also shows technical data for the device:

- device ReSPR serial number*;
- device ID*;
- Location*;
- commissioning date*;
- bus voltage;
- shunt voltage;
- current;
- relay status;
- lamp validation.

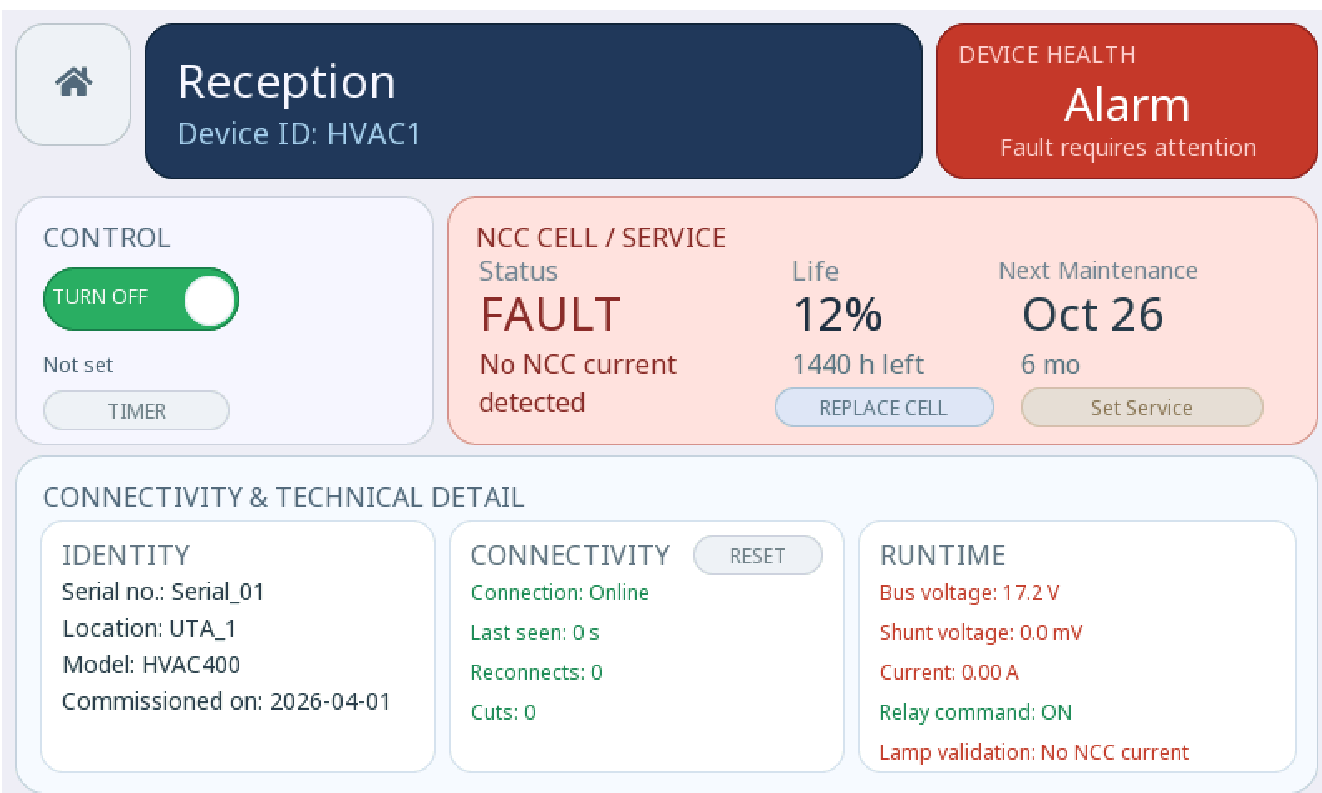
Fields marked with an asterisk (*) are configured by ReSPR Technologies before delivery based on the information provided by the customer within the project framework or, failing that, by their distributor.

Technical values are shown with reference colors:

Color	Meaning
Green	Reading within the expected range.
Yellow	Reading in warning or below target.
Red	Reading in alarm.
Gray	No fresh data or not applicable.



[SCREENSHOT 11 - Device details with warning]



[SCREENSHOT 12 - Device details with alarm]

Restart node

The **RESTART NODE** action is available from the health summary or detail view. This action sends a software restart command to the selected node.

> **Important: RESTART NODE** restarts the selected node, not the HMI. HMI restart is available on Screen 0 and from the dashboard i button. Restarting the node preserves the connectivity history recorded by the HMI.

8. Timers

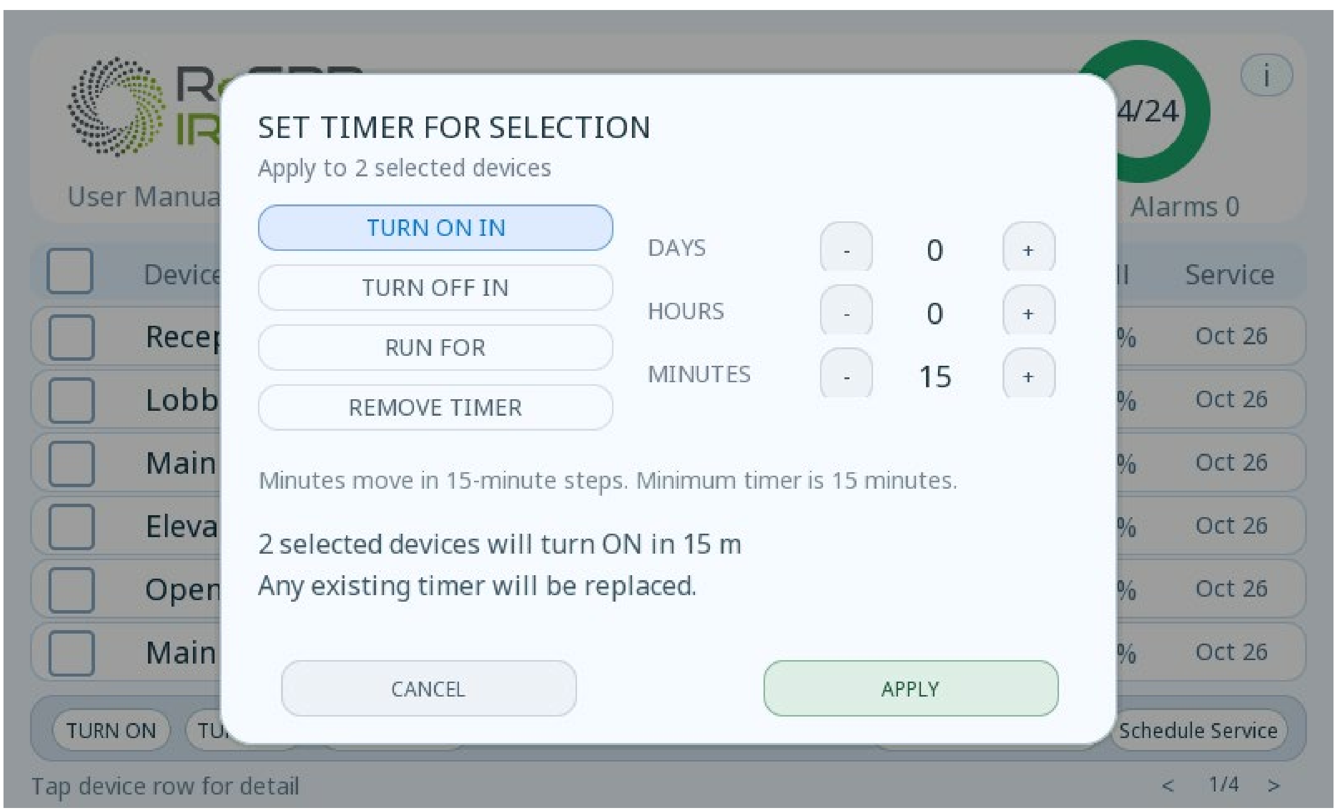
The HMI allows timers to be programmed for one or more selected units.

The timer is set using days, hours, and minutes. Minutes advance in 15-minute steps. The minimum time is 15 minutes. The maximum visible range in the current configuration is 7 days, 23 hours, and 45 minutes.

Applying a new timer replaces any previous timer for the unit.

Available modes

Mode	What it does
TURN ON IN	The device will turn on after the selected time.
TURN OFF IN	The device will turn off after the selected time.
RUN FOR	The device turns on now and will turn off when the specified duration ends.
REMOVE TIMER	Removes the pending timer and keeps the current relay status.



[SCREENSHOT 13 - Timer modal]

TURN ON IN

TURN ON IN mode schedules an ON command at the end of the selected duration. The device remains on indefinitely until a new command is issued.

TURN OFF IN

TURN OFF IN mode keeps the current unit status and schedules an OFF command at the end of the specified time.

RUN FOR

RUN FOR mode sends an ON command immediately and schedules an OFF command at the end of the selected duration.

If the node (ReSPR device) loses power or communication during the active window, the HMI maintains the ON intention as long as the shutoff deadline has not expired. If the node returns before the end, the HMI attempts to restore ON. If it returns after the end, it should not turn it on because the period has already expired.

> **Important**: the duration starts when the user programs the timer, not when the node comes back online.

Timers on offline devices

If a timer is programmed for offline units:

- the timer is saved in the HMI;
- if the deadline expires while the unit is still offline, the HMI will show the status as pending or warning;
- when the unit comes back online, the HMI will attempt to send the command;
- the timer is not cleared until a fresh status is received confirming that the relay reached the target.

For **TURN ON IN**, if the deadline expires while the node is offline, the action will run late when the node becomes available again.

Timers after restarting the HMI

Timers are saved in the HMI's persistent memory. If the HMI restarts:

- a future timer is restored with the saved remaining time;
- an expired or sending timer is restored as pending;
- a confirmed timer is cleared from active memory and persistence;
- a cancelled timer is cleared from active memory and persistence.

> **Note**: exact timing accuracy after a complete loss of HMI power over long periods must be validated in the final installation, because the HMI stores the remaining time and not an absolute external clock time.

9. Meaning of statuses

Online / Offline

Online means that the HMI has received a recent status from the unit. In the current configuration, the freshness window is approximately 20 seconds.

Offline means that there is no recent status. This may be due to unit being switched off, lack of power, loss of mesh communication, insufficient coverage, incomplete startup, or a node issue.

Good / Warning / Alarm

Status	Meaning
Good	No active fault detected, or the unit is in a non-critical condition.
Warning	There is a condition that requires review, such as low current or a pending timer.
Alarm	There is a priority condition, such as low bus voltage, relay ON without active NCC™ current, or loss of connection with last status ON.

NCC™ cell/lamp statuses

Status	Meaning
ON	On the dashboard, indicates active cell with current within the expected range.
FAULT	On the dashboard, indicates that the relay is on, but NCC™ current is insufficient or not detected.
ACTIVE	On Screen 2, indicates an active and healthy cell.
LOW	On Screen 2, indicates current below the expected level, but with activity detected.
FAULT	On Screen 2, indicates absence of NCC™ current when the relay should be on.
STARTING	The device has just received a turn-on command and the HMI is waiting for an NCC™ current reading within the grace period.
INACTIVE	No power applied to the cell.

10. What to do if...

What to do if a unit appears offline

1. Check that the physical device has power.
2. Check that the node or controller has completed startup.
3. Check whether the unit is within mesh coverage.
4. Wait a few minutes if the device has just been energized.
5. If the unit remains offline and other units are online, contact support or review the node installation/coverage.
6. If no unit is online, use the help on Screen 0 or the dashboard i button to restart the HMI if appropriate.

What to do if a warning appears

1. Open Screen 2 for the device.
2. Review the health subtitle and technical details.
3. If the warning is due to a pending timer, confirm whether the unit was offline and wait for it to come back online.
4. If the warning is due to low current, schedule a review of the cell, power supply, or installation.
5. If the warning persists, contact ReSPR support.

What to do if an alarm appears

1. Open Screen 2 for the device.
2. Check whether the alarm is due to a lost connection with last status ON, low bus voltage, or lack of NCC™ current.
3. Physically verify the power supply, node, and cell/lamp.
4. If the device was on and needs to be physically disconnected, first turn it off from the HMI and wait for OFF confirmation whenever possible.
5. Contact ReSPR support if the alarm is not resolved.

What to do if a unit does not respond

1. Wait for a new status update.
2. Check coverage and power.
3. If the device appears online but does not change status, open Screen 2 and review the relay, current, bus, and last signal.
4. Use **RESTART NODE** only if it is the selected node and the installation allows a software restart.
5. If the device still does not respond after restarting the node, contact support.

What to do if several nodes are lost

1. Check the common power supply for the area.
2. Check whether any structural node or repeater has lost power.
3. Check the physical location of the HMI and mesh coverage.
4. If the HMI has gone several minutes without recovering statuses, restart the HMI from Screen 0 or from the i button.
5. Contact ReSPR support if the network does not recover.

What to do if service is pending

1. Open Screen 2 for the device.
2. Review the next service and cell life.
3. Schedule the visit according to the defined interval.
4. After completing service, use the set service action to record the next date and interval.

What to do if an NCC™ cell/lamp has been replaced

1. Physically replace the NCC™ cell or lamp according to the applicable technical procedure.
2. In the HMI, select the device and press **REPLACE CELL** or **REPLACE CELLS**.
3. Confirm only if the physical replacement has already been performed.
4. If the device is offline, the action will be queued and synchronized when it comes back online.
5. Check that **Done** confirmation appears or that life returns to consistent values.

What to do if the screen behaves abnormally

1. Touch the screen to exit dimming or sleep mode.
2. Wait a few seconds if the HMI is processing a reconnection.
3. If the screen does not recover data or the network still has no statuses, use **Restart HMI**.
4. If the problem repeats, contact ReSPR support and provide the device, screen, visible status, and approximate time.

11. FAQ

The HMI remains on "Searching for online devices". What should I do?

Check that at least one HVAC node is powered, within coverage, and correctly configured. Use the startup help to restart the HMI if the search takes too long.

A unit appears offline, but it is physically powered on. Why?

There may be a loss of mesh communication, insufficient coverage, or the node may still be starting up. Check power, distance, and possible obstacles. If other units are online, the issue may be localized to that unit or area.

Why does an offline unit appear as an alarm?

Because the last confirmed status for the unit was ON. The HMI treats this as a priority condition because it cannot confirm that the unit is still operating correctly.

What does it mean when a timer is pending?

It means that the timer expired or that the command was sent, but the HMI has not yet received a fresh status confirming that the relay reached the target. If the node is offline, the HMI will retry when it becomes available again.

How do I cancel a timer?

Open **TIMER**, select **REMOVE TIMER**, and press **APPLY**. This removes the pending action and keeps the relay's current status.

I replaced a cell and the HMI shows "Queued" or "Sync". What does this mean?

Queued means that the action is waiting to be sent, usually because the node is not online. **Sync** means that the command was sent and the HMI is waiting for confirmation. If it does not change to **Done**, check the node connection.

Does restarting the HMI erase data?

According to the expected behavior, restarting the HMI does not modify nodes or erase counters. Timers, service, connectivity counters, and the cell reset queue are saved persistently. Complete power-loss scenarios must be validated in the final installation.

Does restarting a node erase data?

RESTART NODE performs a software restart of the selected node. The HMI connectivity history is preserved. The node's internal persistence depends on the node firmware. In the current relay, lamp hours are saved periodically to reduce memory wear.

Can I use "REPLACE CELL" before physically replacing it?

No. This action must be used only after physically replacing the NCC™ cell or lamp. Using it beforehand may leave the cell life counter in a state that is not consistent with the real installation.

Does the HMI replace an on-site technical inspection?

No. The HMI helps monitor and diagnose the installation, but it does not replace physical checks of power supply, cell, connection, location, service, or operational safety.

12. Operating recommendations

Physical location of the HMI

Install the HMI in an area with good coverage toward the mesh network. Avoid closed metal cabinets, strong structural barriers, or locations that are too far from the nodes or repeaters.

Mesh coverage

The mesh network needs powered nodes with sufficient coverage. In large installations, some nodes or repeaters may act as a communication bridge. The loss of a structural node can affect several units.

Each physical node must have a unique ID that matches the project configuration. Duplicate IDs can cause confusing statuses, commands being sent to the wrong unit, or difficult diagnostics.

When to manually restart the HMI

Manually restart the HMI when:

- the initial screen keeps searching for devices even though there are powered and nearby nodes;
- several nodes remain offline after several minutes and a mesh recovery issue is suspected;
- the HMI shows persistent abnormal behavior;
- ReSPR support requests it during diagnostics.

> **Important**: do not use HMI restart as the first response to a cell or power alarm. First review the affected device.

When to contact your distributor / ReSPR support

Contact ReSPR support if:

- an alarm persists after checking power, cell, and connection;
- online device does not respond to commands;
- several units lose connection repeatedly;
- a cell replacement remains **Queued** or **Sync** for too long;
- the screen requires frequent restarts;
- there are questions about ID, node, or mesh coverage configuration.

13. Final usage note

The ReSPR IRIS HMI interface makes local monitoring of the installation easier and helps detect communication, cell status, timer, and service issues. To maintain safe and consistent operation, critical actions - such as cell replacement, node restart, or alarm review - must be performed following the applicable technical procedure and, when necessary, with support from ReSPR.

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